

BUS RESERVATION AND MANAGEMENT SYSTEM

Programming Fundamentals Term Project Report

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1. Introduction

The Bus Reservation and Management System is a console-based C++ application developed to automate and simplify the process of booking bus tickets and managing transport-related operations. The system allows users to register accounts, log in securely, search available routes, reserve seats, cancel bookings, and view booking information through a token-based mechanism.

The project also provides an administrative panel through which buses can be added, edited, or removed. The system uses file handling to permanently store records such as user accounts, booking details, bus information, reviews, and activity logs.

2. Objectives of the Project

- Develop a complete bus reservation system using C++
- Provide secure user registration and login functionality
- Allow users to book and cancel bus seats
- Maintain records using file handling
- Implement an admin panel for bus management
- Generate booking tokens for users
- Apply sorting and searching functionalities
- Improve understanding of Programming Fundamentals concepts

3. Problem Statement

Manual bus reservation systems are time-consuming and often lead to errors in seat management and booking records. This project solves that problem by creating a computerized reservation system that efficiently handles booking, route management, and user records in a secure and organized manner.

4. Tools and Technologies Used

Tool / Technology	Purpose
C++	Main programming language
iostream	Input and output operations
fstream	File handling and data storage
String Library	String manipulation
ctime	Date and time logging
Text Files	Storage of accounts, bookings, and logs

5. Programming Fundamentals Concepts Used

- Functions
- Arrays
- Structures (struct)

- Loops (for, while, do-while)
- Conditional Statements
- File Handling
- String Manipulation
- Switching and Menu-driven Programming
- Dynamic Memory Allocation
- Searching and Sorting Algorithms

6. Main Features of the System

- User Registration and Login System
- Admin Login and Management Panel
- Bus Addition, Removal, and Editing
- Seat Booking with Seat Map
- Booking Cancellation
- Route Searching
- Route Sorting by Price or Route Name
- Review and Feedback System
- Password Security Validation
- Activity Logging
- Booking Token Generation
- View Personal Booking History

7. System Modules

Registration Module: Allows users to create accounts with strong password validation.

Login Module: Authenticates users and limits incorrect login attempts.

Bus Management Module: Allows the admin to add, edit, and remove bus details.

Seat Reservation Module: Displays seat maps and allows users to reserve available seats.

Booking Management Module: Stores booking details and generates unique booking tokens.

Search and Sorting Module: Allows users to search routes and sort buses according to price or route.

Review Module: Enables users to enter and view reviews.

File Handling Module: Stores all system records permanently in text files.

8. File Handling in the Project

The system uses multiple text files to store data permanently. This allows the application to maintain records even after the program closes.

- **accounts.txt:** Stores usernames and passwords
- **buses.txt:** Stores bus details
- **seats.txt:** Stores seat booking information
- **activity_log.txt:** Stores user and admin activities
- **reviews.txt:** Stores user reviews
- **admin_log.txt:** Stores admin account details

9. Security Features

- Strong password validation
- Limited login attempts
- Unique booking token generation
- Separate admin login system
- User verification before password changes

10. Program Flow

Start Program



Main Menu



Register / Login



Access User or Admin Menu



Book Seat / Search Route / Manage Buses



Store Records in Files



Generate Booking Token



Exit Program

11. Sample Outputs

Example Bus Details Display:

BusNo	Type	Seats	Route	Price
B101	AC	40	Lahore-Karachi	Rs.4500

Example Booking Confirmation:

BOOKING CONFIRMED!

Token: TKN-Abdullah-B101-12

Seat: 12

Departure: 08:00AM

12. Advantages of the System

- Easy to use
- Provides organized booking management
- Reduces manual errors
- Maintains permanent records
- Provides security through password validation
- Fast seat reservation process

13. Limitations

- Console-based interface only
- No database integration
- No online payment system
- Limited graphical interaction
- Works on local text-file storage only

14. Future Improvements

- Develop a graphical user interface (GUI)
- Add online payment integration
- Use a database instead of text files
- Add mobile application support
- Implement real-time seat tracking

15. Conclusion

The Bus Reservation and Management System is an efficient Programming Fundamentals project that demonstrates the practical implementation of core C++ concepts. The project combines user management, booking systems, file handling, searching, sorting, and security features into a single application. It provides valuable learning experience in software development and demonstrates how programming concepts can be applied to solve real-world problems.